

sources_1\new\PONG_BASYS3.vhd

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4
5  --*****
6  --*
7  --* Name: PONG_BASYS3
8  --* Designer: Matthew Mwangi
9  --*
10 --*   This is a wrapper that combines the CLOCK, 7SEG_DISPLAY and the STATE_MANAGER
11 --*   components and maps their input and outputs to their corresponding Basys3
12 --*   board's input and outputs.
13 --*****
14
15
16 entity PONG_BASYS3 is
17 Port(
18     clk: in std_logic;
19     btnC: in std_logic;    --reset
20     btnR: in std_logic;    --player two
21     btnL: in std_logic;    --player one
22     led: out std_logic_vector(15 downto 0); --game board
23     seg: out std_logic_vector(6 downto 0);  --score display (uncomment for advanced pong)
24     an : out std_logic_vector(3 downto 0)    --uncomment for advanced pong
25 );
26 end PONG_BASYS3;
27
28 architecture PONG_BASYS3_ARCH of PONG_BASYS3 is
29
30 component CLOCK      --Incharge of creating the clkTick signal for a 10Hz game play speed
31 Port(
32     reset: in std_logic;
33     clock: in std_logic;
34     clkTick: out std_logic
35 );
36 end component;
37
38 component INPUT_SYNC

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39     Port (  
40         reset: in std_logic;  
41         clock: in std_logic;  
42         aSyncIn: in std_logic;  
43         syncOut: out std_logic  
44     );  
45 end component;  
46  
47 component RESET_SYNC  
48     Port(  
49         reset: in std_logic;  
50         clock: in std_logic;  
51         synchedReset: out std_logic  
52     );  
53 end component;  
54  
55 component LED_DISPLAY  
56     Port(  
57         pos: in std_logic_vector(3 downto 0);  
58         clock: in std_logic;  
59         reset: in std_logic;  
60  
61         leds: out std_logic_vector(15 downto 0)  
62     );  
63 end component;  
64  
65 component SCORE_TRACKER  
66     Port(  
67         scoreSigP1: in std_logic;  
68         scoreSigP2: in std_logic;  
69  
70         reset: in std_logic;  
71         clock: in std_logic;  
72  
73         gameOver: out std_logic;  
74         score: out std_logic_vector(5 downto 0)  
75     );  
76 end component;  
77  
78 component STATE_MANAGER --Handles all the State changes and input synchronizations needed for the game  
79     Port(  
80         p1Btn: in std_logic;
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81         p2Btn: in std_logic;
82
83         reset:   in std_logic;
84         clock:   in std_logic;
85         clkTick: in std_logic;
86
87         ballPos:   out std_logic_vector(3 downto 0);
88         scoreSigP1: out std_logic;
89         scoreSigP2: out std_logic
90
91     );
92 end component;
93
94 component SEG7_DISPLAY --Takes care of displaying the player's score's on the 7seg
95 Port(
96     reset: in std_logic;
97     clock: in std_logic;
98
99     score: in std_logic_vector(5 downto 0);
100
101     anodes: out std_logic_vector(3 downto 0);
102     seg7   : out std_logic_vector(6 downto 0)
103 );
104 end component;
105
106 signal clkTick : std_logic;
107 signal score    : std_logic_vector(5 downto 0); --uncomment for advanced pong
108 signal synchedReset: std_logic;
109
110 signal p1Btn: std_logic;
111 signal p2Btn: std_logic;
112 signal gameOver: std_logic;
113 signal specialReset: std_logic;
114
115 signal scoreSigP1: std_logic;
116 signal scoreSigP2: std_logic;
117 signal ballPos: std_logic_vector(3 downto 0);
118
119 begin
120     -----BLOCK G-----
121     CLCK_BLOCK: CLOCK port map(
122         reset => btnC,
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123     clock => clk,
124     clkTick => clkTick
125 );
126
127 ----BLOCK A-----
128 --*Handles synching the reset btn to the clock
129 RESET_SYNC_BLOCK: RESET_SYNC port map(
130     reset => btnC,
131     clock => clk,
132     synchedReset => synchedReset
133 );
134
135
136 ----BLOCK B SYNC_CHAIN -----
137 --*Handles synching player 1's btn to the clock
138 SYNC_BLOCK_P1: INPUT_SYNC port map(
139     reset => synchedReset,
140     clock => clk,
141     aSyncIn => btnL,
142     syncOut => p1Btn
143 );
144
145 ----BLOCK B SYNC_CHAIN -----
146 --*Handles synching player 2's btn to the clock
147 SYNC_BLOCK_P2: INPUT_SYNC port map(
148     reset => synchedReset,
149     clock => clk,
150     aSyncIn => btnR,
151     syncOut => p2Btn
152 );
153
154 ----BLOCK Y LED_DISPLAY_BLOCK -----
155 LED_DISPLAY_BLOCK: LED_DISPLAY port map(
156     pos => ballPos,
157     clock => clkTick,
158     reset => synchedReset,
159
160     leds => led
161 );
162
163 specialReset <= synchedReset or gameOver;
164
```

```
165 ----BLOCK Z LED_DISPLAY_BLOCK -----
166 SCORE_TRACKER_Block: SCORE_TRACKER port map(
167     scoreSigP1 => scoreSigP1,
168     scoreSigP2 => scoreSigP2,
169
170     reset => specialReset,
171     clock => clk,
172
173     gameOver => gameOver,
174     score => score
175 );
176
177 -----BLOCK C-----
178 SM_BLOCK: STATE_MANAGER port map(
179     p1Btn => p1Btn,
180     p2Btn => p2Btn,
181
182     reset => specialReset,
183     clock => clk,
184     clkTick => clkTick,
185
186     ballPos => ballPos,
187     scoreSigP1 => scoreSigP1,
188     scoreSigP2 => scoreSigP2
189 --     syncReset => syncReset, --uncomment for advanced pong
190 --     score => score,          --uncomment for advanced pong
191 --     led => led
192 );
193
194 -----BLOCK H-----
195 SCORE_DISPLAY: SEG7_DISPLAY port map(
196     reset => synchedReset,
197     clock => clk,
198
199     score => score,
200
201     anodes => an,
202     seg7   => seg
203 );
204
205 end PONG_BASYS3_ARCH;
206
```